Shenova Davis

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EDUCATION

Columbia University **New York, NY**
Master of Science in Data Science, GPA: 3.7 *Dec 2025*
- **Relevant Coursework:** Applied Machine Learning, Deep Learning for NLP, Statistics, Time Series Forecasting, Finance
- **Activities**: Data Science Institute Student Council - Social Chair

University of California, San Diego **La Jolla, CA**
Bachelor of Science in Cognitive Science w/ Machine Learning, Minor in Business, GPA: 3.7 *Mar 2024*

TECHNICAL SKILLS

Languages: Python, SQL, NoSQL, R, Java
Frameworks & Tools: Scikit-learn, PyTorch, TensorFlow, NLTK, Hugging Face Transformers, Apache Spark (PySpark), Snowflake, Palantir Foundry, Docker, Git, Jupyter, SAS JMP, Hadoop, Oracle Financials
Cloud & Visualization: AWS, GCP, Azure, Tableau, Power BI, Excel (Pivot Tables, VLOOKUP, Macros)
Machine Learning: Regression, Classification, Forecasting, Time Series, Natural Language Processing, Clustering, Model Evaluation, Feature Engineering, Reinforcement Learning, A/B Testing, XGBoost, Recommender Systems, Anomaly Detection

PROFESSIONAL EXPERIENCE

Columbia University Irving Medical Center **New York, NY**
Graduate Computational Research Assistant *Sept 2025 – Present*
- Engineered Python-based workflows to retrieve, preprocess, and organize large volumes of biomedical text data from PubMed and PMC, forming the foundation of a structured research database.
- Applied LLM-based text modeling to extract structured features from full-text research papers and integrate them with single-cell RNA-seq and imaging datasets to populate and support analysis within a database.

Pacific Gas and Electric Company (PG&E) **Oakland, CA**
Data Scientist Intern *Jun 2025 – Aug 2025*
- Quantified grid reliability drivers by performing exploratory data analysis (EDA) on 1M+ records and 200+ features using Palantir Foundry, Python, and SQL, identifying the top 5 predictors of circuit failure.
- Developed a Python ensemble forecasting model to identify high-risk circuits with uncertainty estimation, enabling stakeholders to make more informed operational decisions.
- Deployed scalable production pipelines for weekly reliability metrics, automating data flows that previously required manual intervention, ensuring 100% data consistency across regional reports.

Student Life Business Office at UC San Diego **La Jolla, CA**
Human Resources Lead Assistant *Aug 2021 – Sept 2024*
- Facilitated recruitment, onboarding workflows, and database management for 800+ student employees in collaboration with HR Coordinators, ensuring accurate record keeping and timely hiring.
- Led training for new employees to ensure smooth integration and adherence to departmental standards.

UC San Diego Business and Financial Services **La Jolla, CA**
Student Financial Associate *Aug 2023 – Apr 2024*
- Redesigned KPI tracking and reporting processes by conducting large-scale Excel and SQL analysis to guide strategic resource allocation through automated trend analysis, resulting in a 50% improvement in operational efficiency.
- Designed interactive Tableau dashboards to visualize financial health across 25+ departments, enabling leadership to identify budget variances 3 times faster than previous methods.

PROJECTS

Personalized Text Generation with LLMs - CiscoAI (Capstone Project)
- Built a multi-layer Mixture-of-Agents architecture using planner-based prompting and model ensembles to improve personalization metrics such as ROUGE-1 and METEOR
- Improved long-form generation quality through iterative prompt engineering and cross-model validation that produced more coherent and contextually aligned outputs

Electricity Consumption Forecasting
- Developed and tuned Facebook Prophet, DeepAR, and Temporal Fusion Transformer models to forecast electricity consumption across various customer clusters, achieving up to 98% accuracy (MAPE as low as 2.02%).
- Built a robust feature engineering pipeline incorporating external regressors (weather, holidays) and rolling-window statistics to capture seasonal consumption shifts.

Credit Card Application Approval Classification
- Implemented machine learning models (Logistic Regression, Support Vector Machine (SVM), Naive Bayes, KNN, Random Forest, Gradient Boosting) using Scikit-Learn for credit card approval prediction, achieving up to 96% accuracy.
- Engineered a robust preprocessing pipeline to perform one-hot encoding, standard scaling, and SMOTE oversampling, addressing extreme class imbalance (0.4% approval rate).